DATA 301
Applied Machine Learning

Spring 2026

Syllabus

Class Schedule: Lecture: T R: 11:00 AM – 12:20 PM (Section 04) – ISC 3348

Lecture: T R: 3:30 PM – 4:50 PM (Section 03) – ISC 3349

Instructor:
Daniel Vasiliu
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Office Hours: W 4:30 PM – 5:30 PM, T R 5:00 – 6:00 pm, and by appointment.

Course Description: Machine learning is the science of automating decisions and inferences based on data. It uses interdisciplinary concepts and techniques such as statistics, linear algebra, optimization, and computer science to create systems that can parse large amounts of data at high speeds and make automated predictions and decisions with or without human supervision. Machine learning is remarkably pervasive, with applications ranging from business intelligence to homeland security, medical analytics to biochemical engineering, environmental science to astrophysics, etc. This course will introduce students to a broad set of machine-learning ideas, models, and algorithms.

Course Objectives:

1. Provide students with a critical understanding of the essential aspects and techniques in machine learning and the use of data.

2. Develop students' ability to apply different machine learning methods and interpret the results.

3. Expose students to real-world problems that are being engaged with by contemporary problem-solvers and decision-makers.

Evaluation:

Class Participation	10%
Homework/Quizzes	50%
Midterm	20%
Final	20%

Letter Grades

93 – 100 %	A	73 – 76.99%	C
90 – 92.99%	A-	70 – 72.99%	C-
87 – 89.99%	B+	67 – 69.99%	D+
83 – 86.99%	B	63 – 66.99%	D
80 – 82.99%	B-	60 – 62.99%	D-
77 – 79.99%	C+	0 – 59.99%	F

Resources: For programming, you need access to reliable cloud computing services, such as Google Colab Pro. We will use Piazza and (optionally) Discord for discussions and class participation. Obtaining access to capable cloud computing hardware for neural network and deep learning applications will be important, so we encourage you to sign up for Google Colab Pro for free using the education link. **Please sign up for the free Colab Pro ASAP, so your credits can**

start building up in preparation for later modules. There have been students in the past who sign up immediately before Module 4 and therefore run out of credits and are unable to complete the assignments in time. I aim to deliver the study material in a self-contained and efficient manner, so no expensive textbook is necessary, though optional ones are listed just in case. In this course, you will engage with fundamental, conceptual ideas and applications.

Class Participation: You will receive full credit for participation ONLY if you actively engage in class discussions, contribute to coding sessions, and provide feedback to your peers, including Piazza posts. This is **not** a guaranteed 100%; in Spring 2025, the average was ~70%. An average of 20 minutes per class will be used for group discussions and active coding. You can work in small groups to understand and debate the class materials, exercises, and code applications; the instructor mediates all the class discussions. Drafts of the code for solving the problems are shared and discussed in class in a round table format; the instructor and all participants in the debate provide feedback.

Homework: Homework assignments will be posted on Blackboard at least 7 days before the due date. The expectation is that you will **work on the assignment as the material is covered**, so do not be surprised if the due dates are relatively close to when we finish covering the material. If you have any questions, please actively participate in the Q&A on Piazza. For homework projects, you will write your code to solve machine-learning problems and get feedback on improving it. The grading of coding questions also involves partial credit. The lowest homework score will be dropped to accommodate an unexpected personal problem, short-term illness, or class absences.

Quizzes: Quizzes will be on paper in-class, and involve short-answer, multiple-choice, and/or multiple-answer questions, as well as explaining code provided to you (optionally using logical diagrams) to determine the answer. You will **not** be allowed a calculator on any quizzes because you will not need a calculator. You will be allowed to bring both sides of one **handwritten** 4"x6" notecard for each quiz with any notes that you want. You may **not** print the content of your notecard out for any quiz. Quizzes will not be cumulative. In addition to the homework drop, the lowest quiz score will be dropped to accommodate an unexpected personal problem, short-term illness, or class absences. Quizzes and homeworks will be weighted equally.

Midterm and Final: A midterm (March 5th) and a final (May 6th for Section 03 and May 8th for Section 04) will include both in-class, closed-book, conceptual questions (structured like a longer and cumulative quiz) as well as take-home coding components (like a homework). They assess your understanding of the foundational concepts and machine learning algorithms covered in the course. You will **not** be allowed a calculator on any in-person exams because you will not need a calculator. You will be allowed to bring both sides of one **handwritten** piece of letter-sized 8.5"x11" paper with any notes that you want. You may **not** print the content of your notesheet out for the midterm or the final. We will collect your notesheet after the test.

Important Dates: Please review the important dates and plan accordingly.

- The add-and-drop deadline is January 31st.
- Spring Break (no classes) March 7th – 15th.
- Midterm Exam - March 5th.
- The withdrawal deadline is March 23rd.
- The last day of classes is May 1st.
- Final Exam - May 6th for Section 03 and May 8th for Section 04 between 2 pm – 5 pm.

Learning Modules:

Module 1	Python Programming Review, Probability, Monte Carlo Simulations
Module 2	Confidence Intervals and Reinforcement Learning
Module 3	Regression/Classification Problems, Support Vector Machines
Module 4	Intro to Neural Networks (with PyTorch) and Deep Learning
Module 5	Convolutional Neural Networks and Grid Search Optimization Algorithms
Module 6	Natural Language Processing

Weekly Schedule:

Week	What we will discuss/do in class	What to read (before class)
Week 1	Probability, Monte Carlo Simulations	Experimental Probability, Bayesian Rule, Probability Distributions
Week 2	Central Limit Theorem, Reinforcement Learning	Confidence Intervals, Markov Decision Processes
Week 3	Reinforcement Learning	Operations with NumPy arrays, finding min or max values.
Week 4	Supervised Machine Learning – Gradient Descent and its Implementations, Regularized Regression	Law of Probable Errors, Optimization, Gradient Descent
Week 5	Support Vector Machines and Classification Problems	Stratified K-Fold Cross-validations, Precision, Recall, Cross-Entropy
Week 6	Midterm	Review the first three modules.
Week 7	Spring Break (March 7 – 15)	
Week 8	Intro to Neural Networks	Multilayer Perceptron in PyTorch Library, Backpropagation
Week 9	Convolutional Neural Networks	Image Convolutions, Convolution Arithmetic for Deep Learning
Week 10	Grid Search Algorithms	Iterators and Python Dictionaries
Week 11	Neural Network Optimization	Class imbalances, Over/underfitting, Various loss functions
Week 12	Natural Language Processing	Bag of Words, Tokenization

Textbooks (Optional): *The Hundred-page Machine Learning Book* – Andriy Burkov
Data Science from Scratch: First Principles with Python - Joel Grus
Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems – Aurélien Géron
Machine Learning, A Probabilistic Perspective – Kevin P. Murphy

Due Dates:

Quiz 1	February 3rd
Quiz 2	February 12th
HW 1	February 13th
Quiz 3	February 24th
Midterm	March 5th
Quiz 4	March 24th
HW 2	April 3rd
Quiz 5	April 7th
Quiz 6	April 21st
HW 3	April 24th
Final	May 5th, May 8th

Extension Policy: Immediately after a homework is due (which will almost always be midnight ET), any late submissions will have a 15% late penalty applied. Every additional 12 hours up to a maximum of 48 hours late will receive an additional 5% penalty, with the maximum deduction being 30% after

48 hours. Barring exceptional circumstances or prior arrangements, any work submitted after this final deadline will not be accepted. You may request an extension ahead of the due date for any reason. If the extension is approved, the late deductions will only apply beginning after the new due date. For extensions due to illness or sudden emergencies, it will be necessary to submit some form of documentation (e.g., a doctor's note) within a week of the request's approval. If documentation is not submitted within 7 days, penalties may vary, but will generally result in a score of 0 on the assignment.

Collaboration: Collaboration is both allowed and encouraged. Most of the time, we do not work in isolation, and using the resources available is both wise and efficient. You are welcome to work collaboratively with classmates – that means you can work together to solve problems together…not that one person does the work and others copy it. You are also welcome to use the internet as a resource to refresh your memory, clarify concepts, and help with short code snippets. You must cite your sources, and you should not be copying anything wholesale from any source: internet, human, artificial intelligence, or otherwise.

Things that are off limits:

- Soliciting (hey, what did you get for question X? Can you send me your code?) and copying code and answers from another person.
- Copying code/answers directly from internet sources.
- Using sources appropriately but failing to cite them.
- Getting generative AI to do your work for you.

Think about it: do you want to do all of the work and then give the answers to someone? Is that fair to you? Do you wholeheartedly trust that the answers you get from sources (ahem, ChatGPT) will be 100% correct? Do you want to attend a job interview and explain that you don't understand the material? Do you know the answers you are deriving from external sources well enough that you could explain them when asked?
You should always be ready and able to defend every answer you submit confidently. This means that if I ask you to meet with me to explain any answer you submit, you should be able to do so in detail yourself.

Possible Changes: Any changes to the following course schedule or due dates will be announced in class and on Blackboard in advance.

Guidance: Students who are performing at a C level or below may schedule a meeting with the instructor to discuss their class performance.

Class Absences: If you miss a class or two, check the material covered in class posted on Blackboard. For additional help, you can make an appointment with the TAs or the instructor.

Honor Code: Academic integrity is at the heart of the university, and we are all responsible for upholding the ideals of honor and integrity. The student-led honor system is responsible for resolving any suspected violations of the Honor Code, and I will report all suspected instances of academic dishonesty to the honor system. The Student Handbook (www.wm.edu/studenthandbook) includes your responsibilities as a student. Your full participation and observance of the Honor Code are expected. To read the Honor Code, see www.wm.edu/honor.

Artificial Intelligence: The use or incorporation of ANY AI-generated content (from ChatGPT, Gemini, Co-Pilot, etc.) in graded assignments or exams is strictly prohibited. This **will** be enforced to the extent possible.

Expectations:

Student Accessibility: William & Mary accommodates students with disabilities in accordance with federal laws and university policy. Any student who feels they may need an accommodation based on the impact of a learning, psychiatric, physical, or chronic health diagnosis should contact Student Accessibility Services staff at 757-221-2512 or sas@wm.edu to determine if accommodations are warranted and to obtain an official letter of accommodation. For more information, please see www.wm.edu/sas.

Student Success: Student Success supports students' personal growth, development, engagement, and belonging by providing holistic guidance as they navigate their W&M journey. It encompasses the offices of Academic Wellbeing, Care Support Services, and Student Accessibility Services.

- For academic support such as tutoring, time management, study skills, and educational coaching, please contact Academic Wellbeing at wm.edu/academicwellbeing (academicwellbeing@wm.edu)
- For concerns about the well-being of a member of the William & Mary community or to seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu)
- For accommodation needs or questions, please contact Student Accessibility Services at wm.edu/sas (sas@wm.edu)

Student Health: William & Mary recognizes that students juggle different responsibilities and can face challenges that make learning difficult. There are many resources available at W&M to help students navigate emotional/psychological, physical/medical, material/accessibility concerns, including:

- The W&M Counseling Center at (757) 221-3620. Services are free and confidential.
- The W&M Health Center at (757) 221-4386.
- To seek assistance for interpersonal, academic, and wellness challenges, please contact Care Support Services at wm.edu/care (care@wm.edu).
- For a list of other resources available to students, see here or:

Students who experience COVID-19 symptoms during the semester must make an appointment with the Student Health Center for a clinical assessment and testing if necessary. Students who test positive after the start of the semester and need to be quarantined must be cleared by the Student Health Center or a physician before returning to their residence hall and classes.

Living off campus? Visit the Off-Campus Isolation Guide.

For psychological/emotional stress, contact the Counseling Center by email or 757-221-3620.
For physical/medical concerns, contact the Health Center by email or at 757-221-4386.
Both are located at 240 Gooch Drive. Services are free and confidential.
If you or someone you know needs additional support or resources, please contact Care Support Services by submitting a care report online, by phone at 757-221-2510, or by email.